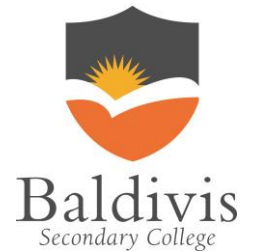


Year 9 - 2015

Investigation 1

Semester One Term 1

Probability



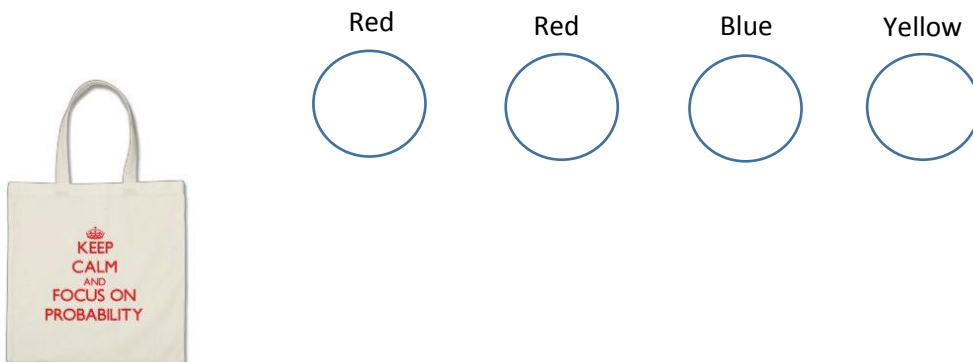
This investigation has three sections. Section 1 will be completed in class as a group. Section 2 is to be completed at home and Section 3 will be completed in class under test conditions. For Sections 1 and 2 you are allowed to ask for help and guidance, however, Section 3 will be completed under test conditions to fully test your understanding. No guidance or hints will be provided.

AIM: *In this assessment you will be investigating probabilities of two step events, calculating relative frequencies both with and without replacement. This will be done using both diagrams and calculations.*

Section 3

Name: _____ Class: _____ Mark: _____ /18

You have a bag which contains 4 coloured balls: two red, one blue and one yellow.



You choose a ball at random from the bag without replacement. You then choose a second ball and record your outcome.

1. If you were to try this experiment, how many times would you need to complete it to make it valid? Why?
(2 marks)

2. Draw a tree diagram to show the theoretical probability for each possible choice combination.

(5 marks)

3. Using your tree diagram, find the following probabilities:

(5 marks)

a. $\Pr(\text{red and blue in any order})$

b. $\Pr(\text{yellow then red})$

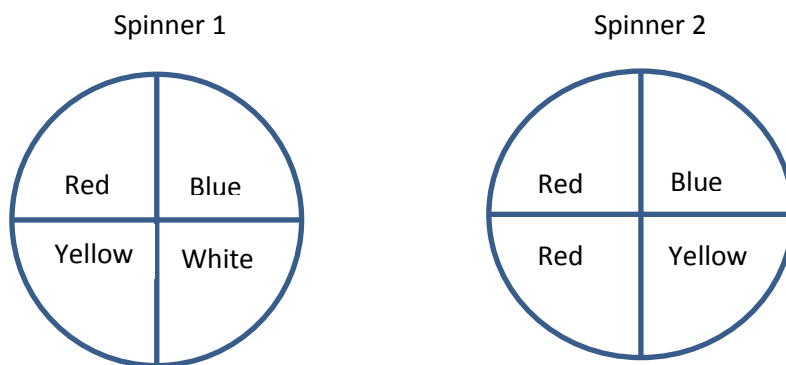
c. $\Pr(\text{not yellow})$

d. $\Pr(\text{not red})$

e. $\Pr(\text{yellow or blue on first choice})$

4. Consider the following frequency table. Is there a difference between the experimental probability (relative frequency) and theoretical probability for this sample space? Why?

(6 marks)



Spinner 1	Spinner 2	Frequency	Relative Frequency
Red	Red	2	
Red	Blue	5	
Red	Yellow	1	
Blue	Red	3	
Blue	Blue	2	
Blue	Yellow	4	
Yellow	Red	0	
Yellow	Blue	1	
Yellow	Yellow	3	
White	Red	2	
White	Blue	4	
White	Yellow	3	

Hint: Use this space to calculate the theoretical probability for this sample space.

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